

Meriem BEN CHAABEN

Computer Science Ph.D Candidate

📍 Montreal, Canada

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🎓 EDUCATION

2023 - ongoing	Ph.D student, Computer Science , At the university of Montreal (UdeM).
2021 - 2023	Computer Science Master student, AI Major with thesis option , At the university of Montreal (UdeM).
2020 - 2021	Information Processing and Complexity of the living Master student , At the National Engineering School of Tunis, Tunisia (ENIT).
2018 - 2021	Software Engineering Student , At the National Institute of Applied Science and Technology, Tunisia.
2016 - 2018	Preparatory cycle , At INSAT, Mathematics, Physics and Computer Science.

🎓 SCHOLARSHIPS

Fall, Winter 2022, 2023, 2024, 2025	Excellence grant from the DIRO (Department of Computer Science and Operations Research) competition , University of Montreal (UdeM).
Winter 2021	International Students Scholarship , As part of the program of the Ministry of Higher Education aiming at attracting international students to graduate programs in engineering and retaining them.
Winter 2021	DIRO Excellence-Writing-Support scholarship competition , University of Montreal (UdeM).
2021 - 2026	TUNISIAN National Scholarship for studies in Canada

📁 WORK/INTERNSHIPS

Sept. 2022 July 2026	Graduate Teaching Assistant, UNIVERSITY OF MONTREAL, DIRO Worked as a teaching assistant in French-taught undergraduate computer science courses by leading lab and demo sessions, explaining technical concepts, assisting instructors, preparing and grading assignments, and holding office hours. ➤ Software Engineering (Summer 2026) — Assisted students with core software engineering practices, including requirements analysis, software architecture, design documentation, version control, testing, and team-based development workflows. ➤ Web Design and Development (Winter 2026) — Guided students through the design and implementation of web applications, covering HTML, CSS, JavaScript, responsive design, client-server interaction, usability, and good development practices. ➤ Software Quality and Metrics (Fall 2022, Fall 2023, Fall 2024, Fall 2025) — Helped students understand software quality assurance concepts, including code quality, testing strategies, software metrics, maintainability, technical debt, and evaluation of software systems. ➤ Programming 2 (Winter 2025) — Led lab sessions focused on intermediate programming concepts such as object-oriented programming, data structures, algorithms, debugging, modular design, and problem-solving in code. ➤ Software Analysis and Design (Winter 2023) — Assisted students with software modeling and design methods, including UML diagrams, requirements specification, use cases, class design, system architecture, and translating user needs into software solutions.
Feb. 2021 June 2021	Automating Software Quality Benchmarking, UNIVERSITY OF MICHIGAN-DEARBORN, ISE Lab, End-of-Studies Internship ➤ Applied clustering techniques to create customized benchmarks for software quality assessment ➤ Estimated operational costs required to achieve targeted quality improvements ➤ Developed a module for software quality benchmarking and target analysis
July 2020 Sept. 2020	Research Intern, UNIVERSITY OF MICHIGAN-DEARBORN, ISE Lab ➤ Defined key concepts related to container smells and anti-patterns ➤ Collected and cleaned data from container-based projects

June 2019	Mobile Development Intern, EXPENSYA, TUNISIA, Internship
Aug. 2019	> Supported the Expensya mobile application lifecycle > Contributed to new expense-management features and wrote unit tests

PUBLICATIONS

Apr. 2026 – Jun. 2026	"Breaking Models to Test the Judge : A Mutation Testing Approach for Semantic Evaluators of Domain Class Diagrams", CONFERENCE PAPER, > Proceedings of the 29th International Conference on Model Driven Engineering Languages and Systems (MODELS 2026), New Ideas and Emerging Results (NIER) Track
Oct. 2024 – June 2025	"On the Utility of Domain Modeling Assistance with Large Language Models", ACM TRANSACTIONS ON SOFTWARE ENGINEERING AND METHODOLOGY, CSS on Human-Centric SE > Journal-first paper in FSE 2026, Montreal, Canada > Accepted in June 2025
Apr. 2024	"Software Modeling Assistance with Large Language Models", ACM STUDENT RESEARCH COMPETITION – DETAILED ABSTRACT AND POSTER, > Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS 2024) > In-person competition : Sept. 25, 2024, Linz, Austria > Award : Second Place
Mar. 2024	"Toward Intelligent Generation of Tailored Concrete Syntax", CONFERENCE PAPER, > Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS 2024) > In-person presentation : Sept. 27, 2024, Linz, Austria
Dec. 2022	"Towards Using Few-Shot Prompt Learning for Automating Model Completion", CONFERENCE PAPER, > International Conference on Software Engineering (ICSE 2023), New Ideas and Emerging Results (NIER) track > In-person presentation : May 17, 2023, Melbourne, Australia

PROJECTS

April. 2026	Agentic AI for Model Generation Across Modeling Formalisms, UNIVERSITY OF MONTREAL, DIRO, GEODES Research Group
Present	> Developed an agentic AI-based approach for generating models across multiple modeling formalisms > Designed workflows where AI agents reason over modeling requirements, constraints, and formalism-specific syntax to produce valid model artifacts
Sept. 2024	Generalizing AI-Assisted Model Completion Across Modeling Formalisms, UNIVERSITY OF MONTREAL, DIRO, GEODES Research Group
Mar. 2026	> Extended an AI-assisted design-completion framework to support multiple modeling languages beyond UML class diagrams > Developed the RAFIC tool by combining semantic reasoning with syntactic constraints to improve model completion quality > Designed and conducted evaluation experiments to assess the tool's effectiveness across different modeling formalisms

June 2022	Automating Model Completion : Designing and Developing a Design Completion Tool, UNIVERSITY OF MONTREAL, DIRO, GEODES Research Group
June 2023	> Applied prompt-engineering techniques to improve completion in domain modeling tasks > Built an editor to assist designers in creating class diagrams > Conducted a user study to evaluate the efficiency and correctness of the approach
Sept. 2021	Learning from Java Code Sources to Suggest Concepts for Class Diagram Completion, UNIVERSITY OF MONTREAL, DIRO, GEODES Research Group
Jan. 2022	> Implemented an iterative diagram-construction process with concept recommendation > Applied MDRE principles to mine Java projects and extract relevant data > Created subgraphs and tuned machine learning models > Evaluated the proposed procedure Python MoDisco GNN MDE Reverse Engineering
Jan. 2021	Classifying Musical Instrument Sounds, ENIT, Academic Project
Feb. 2021	> Applied deep learning techniques to classify instrumental sounds and analyze hyperparameters Python Librosa Keras TensorFlow
Nov. 2020	Tweets Mood Project, INSAT, Academic Project
Dec. 2020	> Designed a big-data project to collect tweets in real time and analyze social mood Kafka Spark PySpark
Feb. 2020	Analyzing and Detecting Fake Reviews of Mobile Apps, INSAT, Final Year Project
June 2020	> Developed a multi-label classifier to assign reviews to representative topics > Defined features to characterize reviews > Built a classifier to detect fake reviews in app stores Keras Scikit-learn Python Google Colab
Nov. 2019	LINK – Secure Web and Desktop Application, INSAT, Academic Project
Jan. 2020	> Developed a web platform connecting sponsors and organizations > Built a desktop chat module secured with LDAP, RSA, and DES LDAP OpenSSL PyQt5 Node.js Angular Python
Jan. 2019	Doctors’ Platform, INSAT, Academic Project
May 2019	> Developed a web platform to support doctor-patient communication and appointment management SCRUM Angular Spring Boot JHipster MySQL

SKILLS/CERTIFICATION

Programming Languages	Java	Python	C/C++	C#	TypeScript	JavaScript
Web Frameworks	Laravel	Spring Boot	.NET	NodeJS	ANGULAR	
SGBD	ORACLE	MySQL	SQL Server			
MDE concepts, tools and languages	EMF	Eclipse Sirius				
Data Science	Keras	Scikit-learn	Tensorflow			
DEEP LEARNING SPECIALIZATION	DEEPLARNING.AI	COURSERA	NLP	CNN		

COMMUNITY INVOLVEMENT

Peer Reviewing

Oct. 2025	Peer Reviewer , Journal of Automated Software Engineering (ASE)
Feb. 2025, Aug. 2026	Peer Reviewer , Journal of Object Technology (JOT)
May 2024; Mar. 2026; Jun 2026	Peer Reviewer , Journal of Software and Systems Modeling (SoSyM)
Aug. 2026	Peer Reviewer , ACM Transactions on Software Engineering and Methodology (TOSEM)
Jul. 2026	PC Member, Double-Blind Peer Reviewer , IEEE International Conference on Embedded & Distributed Systems (EDiS 2026)

Student Volunteering

July, 2026	Student Volunteer , International Conference on the Foundations of Software Engineering (FSE)
Apr. 2025	Student Volunteer , International Conference on Software Engineering (ICSE)
Mar. 2025	Student Volunteer , IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)
Oct. 2022; Oct. 2024	Student Volunteer , MODELS

Talks, Demos & Outreach

May 22, 2026	Poster Presentation , DIRO Research Day (English), presenting results from <i>Generalizing AI-Assisted Model Completion Across Modeling Formalisms, RAFiC Tool</i>
May 23, 2025	Poster Presentation , DIRO Research Day (French), presenting results from <i>On the Utility of Using Large Language Models for Domain Modeling</i>
Apr. 24, 2025	Poster Presentation , Software Engineering at Montreal (SEMTL) (English), presenting results from <i>On the Utility of Using Large Language Models for Domain Modeling</i>
Oct. 18, 2024	Research Presentation , SEMTL, presenting key results from <i>Utility of Domain Modeling Assistance with LLMs</i>
Jan. 31, 2024	Research Demonstration , Expert Network for Model-Driven Engineering (MDENet)
Apr. 2024, 2026	Participant , Journée Francophone pour les Femmes en Informatique (JFFI)
Oct. 20–21, 2023	Lightning Talk , Annual Canadian Celebration of Women in Computing (CAN-CWiC)
Apr. 2023	Speaker , Women in Computer Science Research Panel, UdeM

Leadership & Clubs

2018–2019	Vice President , INSAT ANDROID Club
2017–2018	Treasurer & Event Sponsorship Lead , IEEE Computer Society Chapter, INSAT

Hackathons & Competitions

Sept. 2024	Student Research Competition – Detailed Abstract & Poster , MODELS 2024 proceedings; in-person competition, Sept. 25, 2024, Linz, Austria. Award : Second Place
Sept. 2019	AI Hack Tunisia
Apr. 20, 2019	Infor Hackathon 4.0. Award : Third Place
Feb. 2017	Contestant, INSAT Collegiate Programming Contest (ICPC)